M4 Light pitch

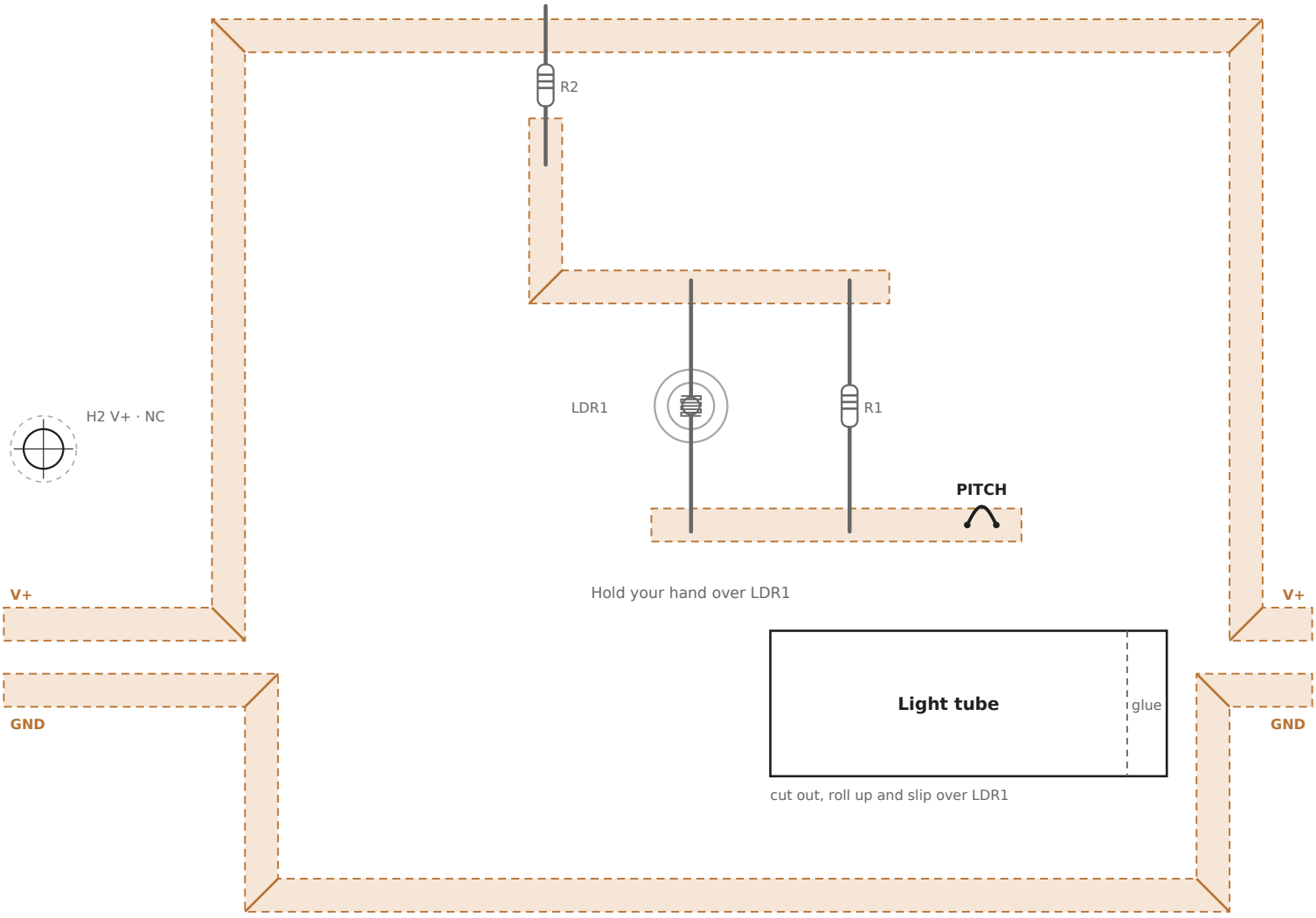
Wave a hand over a light sensor, and the note follows.

Draft: tagged steps may change

H1 BUS A · NC

You need

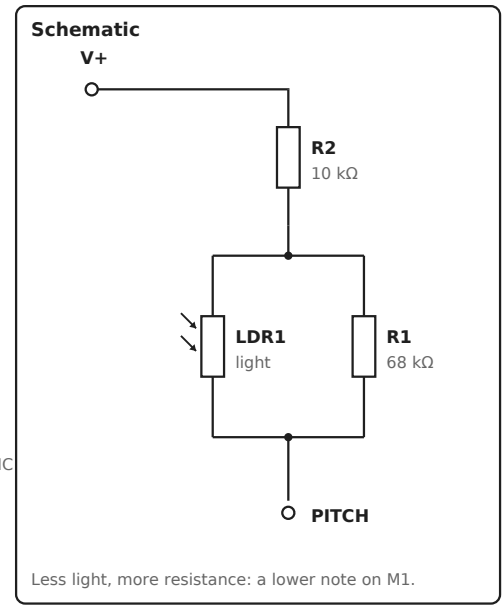
- LDR1: 5 mm LDR, such as a GL5528
- R1 68 kΩ · R2 10 kΩ
- Bare wire for 1 patch loop
- A clip lead to M1's PITCH
- A desk lamp, for the widest range
- Copper tape, solder, 2-core cable



H2 V+ · NC

H3 GND · NC

Patch PITCH to M1's PITCH. Shade LDR1 for a lower note. Only V+ is used; GND passes through.



H4 BUS B · NC

Build it

1. Lay the power ring and the two lanes inside it, one piece of tape each.
2. Bend LDR1's legs flat in opposite directions, so its face looks up, and place it across the two lanes. Either way round.
3. Place R1 beside it, and R2 from the top rail to the upper lane. Either way round.
4. Solder every lead to its lane, or hold it flat with a small patch of copper tape. **[T0]**
5. Solder a patch loop upright on the PITCH loop mark. **[T0]**
6. Join the edge connectors to the pages either side with 2-core cable, or with red and black clip leads.

Play it

- Patch PITCH to M1's PITCH. Hold a hand over LDR1 and lower it: the darker, the lower the note.
- Cut out the light tube, roll it round a pencil, glue the flap and slip it over LDR1. Now point it at a lamp to play.
- Take R1 out, and a full shadow fades the note to silence.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

Print at 100 %: this bar should measure 100 mm.